# Jean Luc BIAH

## Tech Lead .NET & React / Technical Project Manager — Finance and Insurance

Freelance Tech Lead with 20 years of experience in banking and insurance. I lead teams and financial and actuarial computing applications (ALM, AFM, HPC), design their architecture and industrialise their delivery (CI/CD, automated deployment). I modernise legacy applications (WinForms, WPF, VB) to .NET Core and React. I build generative AI (Anthropic API, coding agents) into my products and team practices.

Paris area, France · Contract (régie), on site or hybrid · Available within 3 months (notice period)

jeanlucbiah.pro@outlook.com · www.jeanlucbiah.com · LinkedIn · GitHub

## SKILLS

| Area | Technologies |
|---|---|
| **Back end** | C#, .NET 8 / .NET Core, ASP.NET Core, ASP.NET MVC, Windows Services, Node.js / Express, Python / FastAPI, REST APIs, SignalR, WCF, RabbitMQ |
| **Front end** | React, Next.js, React Native / Expo, JavaScript, Vue.js, Blazor (WebAssembly), WPF, Tailwind CSS |
| **Data** | SQL Server, PostgreSQL, Oracle, MongoDB, T-SQL, PL/SQL, SSIS, SSAS, SSRS, Dapper, Entity Framework |
| **DevOps and cloud** | PowerShell, IIS, NuGet, Artifactory, GitLab CI, Azure, Docker, Kubernetes, Ansible, Jenkins, TeamCity, Octopus, Vercel, Railway |
| **Quality and observability** | NUnit, Moq, TDD, SonarQube, Datadog, Kibana, Graylog, Sentry |
| **Financial computing** | ALM and AFM computing grids, MS HPC, distributed computing orchestration (HpcLite), DAG dependencies, real-time job monitoring |
| **AI** | LLM integration (Anthropic API, OpenAI), AI-driven automation |
| **Methods** | Scrum, Kanban, Epic / Feature / User Story backlog, Jira, Software Craftsmanship, SOLID, design patterns (GoF), UML |

**Business domains:** life insurance and ALM, actuarial science, Forex, counterparty risk, guarantees and sureties, Solvency II, US Treasury bonds.

## EXPERIENCE

### Sogecap (Société Générale Insurance), Paris — Application Officer, Technical Project Manager / Tech Lead (employee)

**Since 03/2021**

Applications that launch and monitor actuarial (ALM) computations, used by actuaries. Agile team of 7, including 3 front-end developers under my lead.

**Computing platform migration (2026), led as Application Officer — POC approved, development nearly complete**

- Designed **HpcLite**, a .NET 8 orchestration layer that reproduces the logic of Microsoft HPC (scheduler, agents and runners as Windows Services) and submits tasks to the Group's internal computing grid.
- Built the POC back end in .NET 8 / Dapper: REST API, job dependencies as a graph (DAG), state machine, crash detection, safe concurrent dispatch on PostgreSQL (`SKIP LOCKED`), integrated with the existing data model.
- Applied least privilege (agents without database access) and documented the target architecture; code skeletons generated with Claude Code from iterative specifications.
- Built the POC front end (HTML, CSS, JavaScript): submission interface, real-time monitoring of computation progress and logs, IndexedDB cache and client-side optimisations, on a design system shared by two applications.
- Beforehand, an exploratory POC in Node.js / Sequelize, which became the sister application whose design system was reused.
- Set up agile delivery: Epic → Feature → User Story backlog, sprints tracked as GitLab milestones, standard Jira ticket template (context, deliverables, acceptance criteria); led the web migration workstream (project plan, demos, acceptance testing, releases).
- Automated end-to-end deployment in PowerShell (Artifactory → IIS and Windows Services, pre-flight checks, rollback), validated through 19 milestones and 167 test scenarios up to production; scripts and documentation handed over to the operations team.
- In progress: GitLab CI/CD for a .NET team not covered by the Group's DevOps platform (Java only). Build script review: removed a hard-coded secret and false-success publications.

**Redesign of the ALM computation launcher**

- Supervised the redesign of the WPF application as a React web application (forms, data tables, dashboards): task breakdown, code reviews, mentoring; maintenance of the WPF version.

- Re-architecture of the solution (GoF, SOLID, Software Craftsmanship); unit tests (NUnit, Moq) with code coverage tracked in SonarQube.
- Scrum delivery and single point of contact for actuaries, software vendors (bugs, licences) and IT (DevOps, servers, HPC).

*Stack: C#, .NET 8, ASP.NET Core, Kestrel, Windows Services, Dapper, PostgreSQL, HTML/CSS/JS, IndexedDB, React, Node.js, Sequelize, MVC, WPF, PowerShell, IIS, NuGet, Artifactory, GitLab, Jira, Jenkins, Ansible, Kubernetes, MS HPC, Claude Code*

### Les Rappelés, Tallinn (Estonia) — Lead Developer (side project)
**03/2025 – 08/2025**

Design and delivery of the Lissen platform, from the mobile app to the back end.

- Expo / React Native mobile app published on Google Play.
- Next.js / React website deployed on Vercel.
- “Forge” back end in Node.js / Express (REST API) on Railway.
- Vercel and Railway CI/CD, crash and performance monitoring (Sentry, Crashlytics).
- AI-driven automation and assistance (Anthropic API, OpenAI).

*Stack: React Native, Expo, Next.js, Node.js, Express, PostgreSQL, IndexedDB, GitHub*

### Ektare Immo, Nairobi (Kenya) — Project Manager (side project)
**06/2023 – 12/2023**

Design and delivery of a property management application for the Kenyan diaspora, with a team of 6.

- Solution design (front end, middle office, back end) and implementation of the middle office and back end.
- MongoDB database and REST API (OpenAPI, Controller / Service / Repository architecture).
- React application for the middle office, showcase website hektare-immo.com (HTML5, CSS3, jQuery).
- Team delivery with Scrum / Kanban.

*Stack: React, Node.js, Express, MongoDB, REST API, Git, GitHub*

### Binas Group, Brussels — Technical Project Manager
**11/2020 – 01/2021**

Greenfield development of a contact management PDM, with a team of 3.

- Solution architecture, SQL Server database and Dapper ORM.
- .NET Core API hosted on Azure, WPF client (Caliburn Micro).

### Sinequanone Institute, Bezons — Master’s degree trainer (M1 and M2)
**Since 08/2020**

Training work-study students in full-stack development, Software Craftsmanship, DevOps (CI/CD) and agile project management. Two platforms delivered: invoicing (MongoDB, Node.js, React, Stripe, Docker) and apprenticeship contracts (Vue.js, Sequelize).

### Betclic Group, Bordeaux — Tech Lead
**11/2019 – 10/2020**

Tech Lead of a group of 4 within a team of 20, redesigning Windows services and REST APIs.

- API receiving real-time match messages from brokers (RabbitMQ).
- Stored procedures, Dapper ORM, Datadog and Kibana monitoring.
- Code reviews and production releases (TeamCity, Jenkins, Octopus).

### Société Générale CIB, La Défense — Lead Developer
**01/2019 – 09/2019**

Maintenance of XOne (Forex Cash): unit test overhaul (NUnit, Moq), PL/SQL procedures on Oracle.

### AG Insurance, Brussels — Lead Developer
**08/2017 – 08/2018**

Evolution of a web application that drives computations on an AFM grid and replaces an Excel tool. WCF back end, ASP.NET MVC screens.

### Candriam Investors Group, Paris — Lead Developer / Architect
**11/2016 – 03/2017**

Migration of VB and VB.NET batches to C# .NET 4: application inventory, dependency analysis, KPIs, test-driven development.

### CEGC (BPCE Group), La Défense — Lead Developer
**10/2014 – 10/2016**

Evolutions of the guarantee and surety management application (ASP.NET, WinForms, WCF, SSRS): technical specifications, architecture, packaging and acceptance testing. Performance audit for BPCE in 07/2014.

### Earlier engagements (2006 – 2014)

- **Banque de France** (2013 – 2014): VSTO Lead Developer on Solvency II, ASP.NET MVC and SSIS.
- **Crédit Agricole CIB** (2011 – 2012): counterparty risk reporting (VaR, P&L, stress), SSIS and SSAS.
- **Société Générale CIB** (2010 – 2011): market connectivity (Tradeweb, Bloomberg) for US Treasury bonds, Scrum pilot project.
- **Natixis** (2009 – 2010): financial product trading applications.
- **Itron, Newedge, C-Design Fashion** (2006 – 2009): industrial software, database administration, PDM.

## GENERATIVE AI

**Fluxo — personal finance management application (2026)**

- Full-stack application designed and delivered on my own: dual-currency budgeting (EUR / XAF), September-to-August fiscal year, bank statement import and reconciliation (CSV, OFX), recurring payments, debts, savings, dashboards, mobile PWA.
- AI financial advisor on the Claude API: streamed answers (SSE), with a system prompt automatically fed with the user's financial context (income, expenses, disposable income, debts, tight months). Prototype removed after evaluation: the value delivered did not justify the API cost. A product decision, not a technical one.
- Built with Claude Code: back-end and front-end foundation (100+ files) up and running in two days, driven by context documentation for the agent (CLAUDE.md, CONTEXT.md) and a development sequence log.
- Stack: Next.js 14 (TypeScript, shadcn/ui, Recharts), FastAPI (Python), SQLAlchemy 2, PostgreSQL 16 (dedicated schema, views, triggers), NextAuth, GitHub Actions CI/CD to Railway and Vercel.

**Bitojel — automatically published quotes media (since 2026)** — x.com/bitojelx

- Automated publishing pipeline: quote scheduling, visual composition (retouched portrait and quote), daily posting on X; Facebook ready, short videos in design. Web control dashboard (import, scheduling, authors, quotes, publication monitoring).
- AI built into the editorial work: the Claude API analyses imports (duplicates, unknown authors, themes), enriches the schedule and writes YouTube metadata; in the author page, a conversational assistant carries out actions itself (biography, watermark).
- Secure Node.js back end: token-based authentication (JWT), Claude API key kept server-side, rate limiting. Built with Claude Code.
- Stack: Node.js / Express, Prisma, PostgreSQL, Sharp, @napi-rs/canvas; Netlify, Railway, Vercel (cron).

**GRIM (2026)** — Electron desktop app (macOS, Windows) to track my AI-built projects: each GitHub repository holds a `summary.md` kept up to date by the agent; GRIM gathers them and reopens the Claude project in one click. React, TypeScript, tests and installers published by GitHub Actions — github.com/jlbiahdev/grim.

**Agent-assisted development** — Claude Code and GitHub Copilot every day, governed by project rules (CLAUDE.md): SOLID principles enforced, no SQL execution without approval.

## INDEPENDENT WORK THROUGH CYRANO CONSEIL

Projects designed and delivered as an independent, alongside client engagements.

| Project | Type | What I delivered |
|---|---|---|
| **Lissen** | Mobile + web | Spiritual-life app built with Expo / React Native and published on Google Play, plus Lissen Forge, a web back office for content and user management |
| **Lyngwa** | API | Data API on countries, cities and islands worldwide, filterable by name, ISO code and currency |
| **Trello Scraper** | API | Trello data extraction API: boards, lists, cards, members |
| **Law For All** | Website | Legal information platform for the general public |
| **Dream Logistics** | Website | Website of a regional training academy for logistics professions |
| **Ektare** | Website | Real estate company website with automated project tracking |
| **Nzuane** | Website | Website of a children's aid charity (Ebolowa, Cameroon) |
| **AutoStock** | E-commerce | Online store for car, motorcycle and truck spare parts |

## EDUCATION AND LANGUAGES

- **Ingesup, Paris** — Engineering degree in information systems and software engineering, 2007
- **Institut Africain d'Informatique, Libreville** — Analyst-programmer, 2003
- **Languages** — French and English, fluent
- **References** — available on request (CEGC/BPCE, Banque de France, ACPR)